

JNANARANJAN SAHOO

AI / Backend Software Engineer

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SUMMARY

Software Engineer specializing in production AI systems — RAG pipelines, real-time voice agents, and computer vision — built on async Python/FastAPI and self-hosted LLM infrastructure. Comfortable owning systems end to end, from multi-tenant SaaS architecture and on-premise deployment to sub-second voice pipelines that run without third-party APIs for data-sensitive clients. AWS Certified in Cloud Practitioner and AI Practitioner.

PROFESSIONAL EXPERIENCE

Software Engineer — Skillblabers Technologies Pvt. Ltd., Bangalore

Jun 2025 – may 2026

- **Real-time voice agent (SurveyBot):** Built a conversational survey agent on self-hosted LiveKit, orchestrating Sarvam and Gemini models across the STT → LLM → TTS pipeline to deliver sub-1-second agent response latency for natural, spoken survey interactions.
- **On-premise RAG application (museum):** Designed and deployed a self-hosted RAG system letting visitors ask questions about the Bhagavad Gita and receive conversational answers; ran the full stack on the client's infrastructure with Ollama-served LLMs (zero external API calls) and held response times under 2 seconds with a caching layer.
- **On-prem deployment & self-hosted AI infra:** Own end-to-end on-premise deployment of client projects — local LLM serving (Ollama behind a LiteLLM proxy) and self-hosted TTS (Kokoro, edge-tts, facebook/mms-tts-ory for Odia) — eliminating paid-API dependence and keeping all client data fully in-house.
- **E-paper digitization platform:** Built a full-stack e-paper solution: a fine-tuned IndicDLP layout-detection model auto-draws article bounding boxes on newspaper pages, article crops are stored in Hetzner object storage with metadata in PostgreSQL, and the frontend renders the boxes so readers can click any article to view it full-size.
- **On-device face recognition (mobile):** Engineered a Flutter face-recognition pipeline — Google ML Kit + face_camera for detection, then a MobileFaceNet TFLite model producing 192-dim L2-normalized embeddings from 112×112 crops, matched by Euclidean distance against a pre-computed embedding store (threshold ≤ 0.9) to identify and route to the correct profile.
- **Computer vision:** Fine-tuned an EfficientNet embedding model for image similarity search (matching live image vectors against a stored index) and built YOLO-based dishonest-object detection for a proctoring / anti-cheating system.
- **Workflow automation & avatars:** Built an LLM-driven email auto-reply generator on n8n + Gemini that first classifies which incoming emails warrant a response, then drafts ~100 replies/day; also built a video avatar agent integrating LiveKit with the Anam avatar provider.
- **AI features for Beyond Resume (skills-based hiring platform):** Built the platform's LLM-driven features — JD generation, interview-question generation conditioned on the JD and candidate profile, adaptive answer evaluation, candidate-JD fitment analysis, ATS optimization, and cover-letter generation — using LangChain's structured-output parser to enforce schema-validated JSON for reliable API consumption..

SELECTED PROJECTS

Swaram — Multi-Tenant Conversational Voice AI Platform

SaaS · in development

Hybrid monorepo (pnpm + uv) · Next.js · React/Vite · async FastAPI · self-hosted LiveKit · PostgreSQL/pgvector · Redis/arq · S3

- **Multi-provider SIP telephony:** Built a bring-your-own-carrier layer — tenants supply their own credentials, buy DID's through the platform, and calls terminate into self-hosted LiveKit SIP. Twilio and Vobiz plug in behind a provider-agnostic interface.
- **Sub-second agent runtime:** Provider-agnostic STT → LLM → TTS pipeline (Sarvam, Gemini) on LiveKit Agents, with pgvector RAG answering caller queries per tenant; engineered to a sub-1-second response-latency budget.
- **Multi-tenant control plane:** Async FastAPI with OAuth + SMS-OTP auth, strict org_id isolation, Razorpay billing, and knowledge-base ingestion — SQLAlchemy/Alembic, Redis (cache, arq, pub/sub), Docker Compose on AWS EC2.

Data Analysis Agent — LangGraph Multi-Tool Agent(MVP)

Freelance · delivered

LangGraph · Gemini · E2B sandbox · FastAPI · PostgreSQL · MinIO · React/Vite · Docker Compose · Nginx · AWS EC2

- Built an autonomous LangGraph agent that profiles uploaded CSV/Excel/JSON, then writes and runs pandas/matplotlib code in isolated E2B sandboxes and streams results with charts rendered inline — backed by dual memory: per-thread PostgresSaver checkpoints and cross-session PostgresStore namespaced per user.
- Designed a security-first deployment: a FastAPI auth proxy (bcrypt + JWT, thread-ownership checks, SlowAPI rate limiting) fronts a LangGraph container with no public port, and MinIO chart storage is reachable only through an Nginx proxy — no internal service is publicly exposed.

- Shipped the full stack on a single AWS EC2 instance via Docker Compose (PostgreSQL, MinIO, LangGraph, FastAPI) behind Nginx with Let's Encrypt SSL.

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI (async), Node.js, Express.js, React, Next.js, Vite, REST APIs, WebSockets / SSE

GenAI & LLM: RAG, LangChain, LangGraph, AI agents, prompt engineering, embeddings, vector search, LLM fine-tuning, self-hosted LLMs (Ollama, LiteLLM)

Voice & Real-time: LiveKit, STT/LLM/TTS pipelines (Sarvam, Gemini, Kokoro, edge-tts, MMS), sub-second latency optimization

Computer Vision: PyTorch, EfficientNet, MobileFaceNet, YOLO, face recognition, image similarity search, document layout detection, on-device / TFLite inference

Architecture & Practices: Multi-tenant SaaS, hybrid monorepos (pnpm / uv), provider-agnostic interfaces, async-first design, OAuth / JWT auth, rate limiting

Databases & Storage: PostgreSQL (SQLAlchemy, Alembic), MySQL, Redis (arq), Qdrant, PGVector, MinIO, S3 / object storage

Cloud, DevOps & Integrations: AWS (EC2), Hetzner, DigitalOcean, Cloudflare, Docker / Docker Compose, Nginx, Linux, Let's Encrypt, n8n, Celery, Razorpay, Twilio

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner

EDUCATION

B.Tech, Computer Science & Engineering — Government College of Engineering, Keonjhar 2020 – 2024